EXPERIMENT 13: FILE ORGANIZATION STRATEGIES

System Environment: kali@Harish | Shell & C Implementation

OS LAB MANUAL - MODULE 13

EXPERIMENT 13-A: Sequential File Organization

C PROGRAM

```
#include <stdio.h>

struct student {
    int regno;
    char name[20];
};

int main() {
    FILE *fp;
    struct student s;

    fp = fopen("student.dat", "w");
    printf("Enter Register Number: ");
    scanf("%d", &s.regno);
    printf("Enter Name: ");
    scanf("%s", s.name);
    fprintf(fp, "%d %s\n", s.regno, s.name);
    fclose(fp);

    fp = fopen("student.dat", "r");
    fscanf(fp, "%d %s", &s.regno, s.name);
    printf("\nRecord Details\n");
    printf("Register Number: %d\n", s.regno);
    printf("Name: %s\n", s.name);
    fclose(fp);

    return 0;
}
```

SHELL SCRIPT (SEQUENTIAL.SH)

```
#!/bin/bash
echo "Enter Register Number:"
read regno
echo "Enter Name:"
read name
echo "$regno $name" > student.txt
echo "Contents of File"
cat student.txt
```

TERMINAL OUTPUT

```
(kali@Harish)-[-]$ nano student.txt
(kali@Harish)-[-]$ nano sequential.sh
(kali@Harish)-[-]$ chmod +x sequential.sh
(kali@Harish)-[-]$ bash sequential.sh
Enter Register Number:
101
Enter Name:
ARUN
Contents of File
101 ARUN
```

EXPERIMENT 13-B: Direct (Random) File Organization

C PROGRAM

```c
#include <stdio.h>

struct student {
    int regno;
    char name[20];
};

int main() {
    FILE *fp;
    struct student s;

    fp = fopen("random.dat", "wb+");
    printf("Enter Register Number: ");
    scanf("%d", &s.regno);
    printf("Enter Name: ");
    scanf("%s", s.name);

    fwrite(&s, sizeof(s), 1, fp);
    rewind(fp);
    fread(&s, sizeof(s), 1, fp);

    printf("\nRecord Found\n");
    printf("Reg No: %d\n", s.regno);
    printf("Name: %s\n", s.name);
    fclose(fp);

    return 0;
}
```

SHELL SCRIPT (RANDOM.SH)

```
#!/bin/bash
echo "Enter Record:"
read rec
echo "$rec" > random.txt
echo "Random Access Record"
sed -n '1p' random.txt
```

TERMINAL OUTPUT

```
(kali@Harish)-[-]$ nano random.txt
(kali@Harish)-[-]$ nano random.sh
(kali@Harish)-[-]$ chmod +x random.sh
(kali@Harish)-[-]$ bash random.sh
Enter Record:
12
Random Access Record
12
```

EXPERIMENT 13-C: Indexed File Organization

C PROGRAM

```c
#include <stdio.h>

struct student {
    int regno;
    char name[20];
};

int main() {
    struct student s[3];
    int key, i;

    printf("Enter 3 Student Records\n");
    for(i = 0; i < 3; i++) {
        scanf("%d %s", &s[i].regno, s[i].name);
    }

    printf("Enter Register Number to Search: ");
    scanf("%d", &key);

    for(i = 0; i < 3; i++) {
        if(s[i].regno == key) {
            printf("\nRecord Found\n");
            printf("Reg No: %d\n", s[i].regno);
            printf("Name: %s\n", s[i].name);
            return 0;
        }
    }

    printf("Record Not Found\n");
    return 0;
}
```

SHELL SCRIPT (INDEX.SH)

```bash
#!/bin/bash
echo "Enter Student Records"
echo "101 Arun" > index.txt
echo "102 Kumar" >> index.txt
echo "103 Ravi" >> index.txt
echo "Enter Register Number to Search:"
read key
grep "^$key" index.txt
```

TERMINAL OUTPUT

```
(kali@Harish)-[-]$ nano index.txt
(kali@Harish)-[-]$ nano index.sh
(kali@Harish)-[-]$ chmod +x index.sh
(kali@Harish)-[-]$ bash index.sh
Enter Student Records
Enter Register Number to Search:
101
101 Arun
```